Youngstown Streetlight Traffic Analysis

Ethan Gerak, Bishal Lamichhane & Joshua Teye

Youngstown State University

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Supervisor: Dr. Jay Kerns

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Abstract

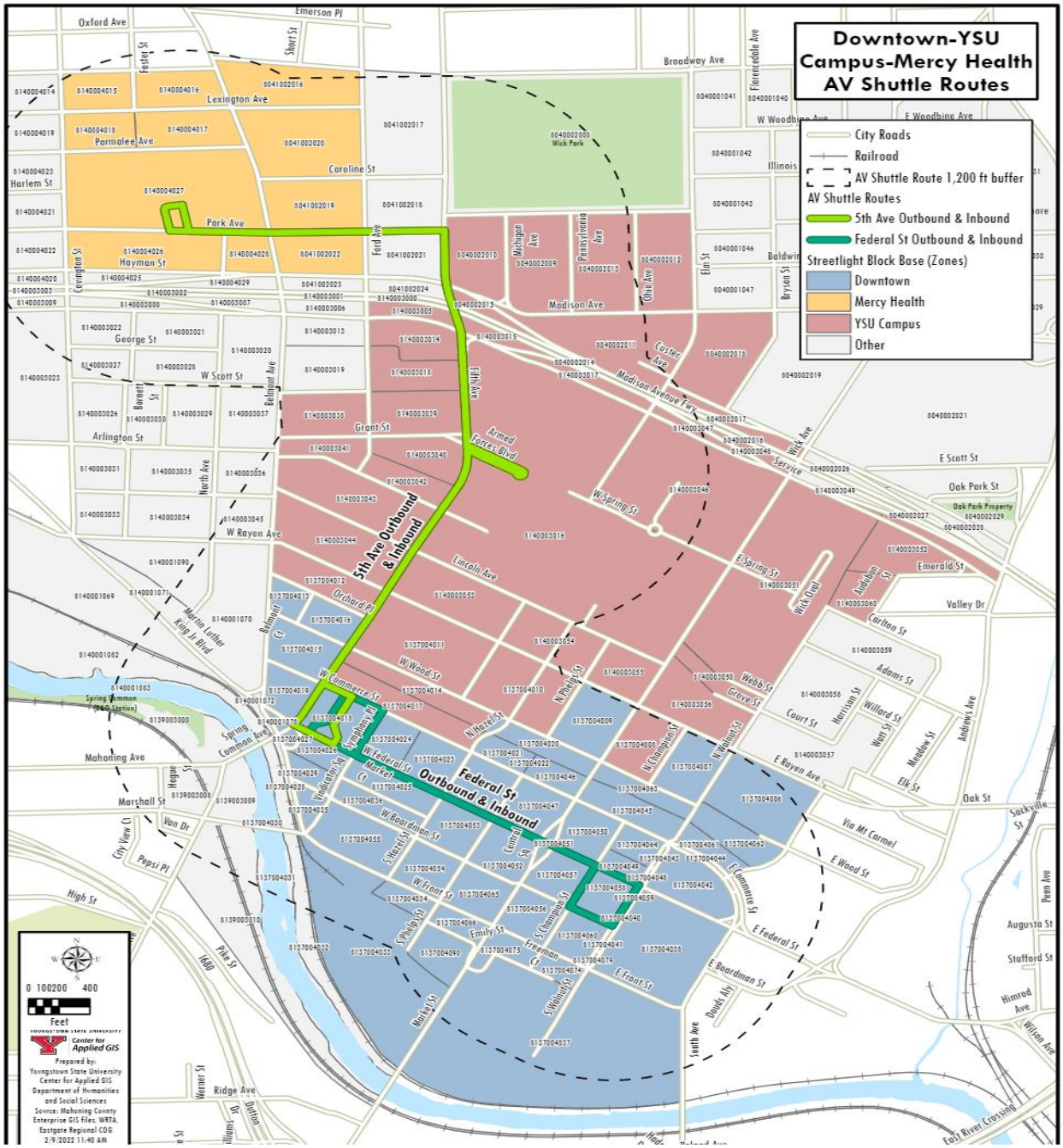
We examine October 2019 traffic patterns throughout the city of Youngstown, Ohio for the purpose of the installation of an autonomous shuttle system. The proposed autonomous shuttle system is planned to run from the Western Reserve Transit Authority's (WRTA) federal station, located on West Commerce Street, up along 5th Avenue and ending at Mercy Health. The primary purpose of this route will be to transport community members across the key regions of Youngstown: Downtown, Youngstown State University, and Mercy Health. The summary statistics reports that the highest individual travelers, categorized by their level of education are the individuals with High School Diploma holders of about 42%, by their race, the White race emerged as the highest travelers of about 70%, by their level of income, the individuals with an income less than $20,000 emerged as the highest travelers of about 31% and by having a kid or not, the individuals with no kids emerged as the highest travelers of about 50%.Our analysis utilized the *Streetlight Data* provided to us through the Eastgate Regional Council of Governments. The dataset captures 194 Census Blocks – the smallest geographic unit used by the United States Census Bureau – and contains demographic data as well as travel data derived from smartphone GPS location tracking captured through smart-phone applications. We analyze traffic patterns through both time and location to identify geographic hotspots across Youngstown and popular travel patterns. Our research is intended to guide city planning decisions in Youngstown, Ohio and utilizes all relevant traffic data available for Youngstown City.

Introduction

Youngstown State University and Eastgate Regional Council of Governments have long partnered to promote the economic prosperity of Youngstown, Ohio. Due to the COVID pandemic, as well as the longstanding crime rate that has plagued Youngstown, travel between University Campus to the downtown area has seen a rapid decline throughout the past decade. Furthermore, Youngstown grapples with a severe poverty rate of 35%, 21% higher than the statewide average. As a result, much of the population relies on public transport for travel to important locations such as Mercy Health, the primary hospital in the area. To aid in this travel, the City of Youngstown was awarded a BUILD grant from the US Department of Transportation for the reconstruction and renovation of the downtown area. Incorporated into the grant was a plan to build and operate an autonomous shuttle in Youngstown. Following the BUILD grant award, Western Reserve Transit Authority was awarded an AIM grant from the Federal Transit Administration to demonstrate a route and how it might be of service to the in-need population of Youngstown.

The initial shuttle route was set to operate along the fifth avenue, connecting downtown Youngstown, Youngstown State University, and Mercy Health. This route would theoretically maximize the number of riders, as the route incorporates all three major zones within Youngstown. However, to ensure that this is the best possible route, Western Reserve Transit Authority plans to change route locations during the second year of operation. During the second year, the autonomous shuttle would operate across Federal Street, located in downtown Youngstown. This route would sacrifice the broad accessibility that the fifth avenue route provides, in exchange for a shorter path, directly adjacent to the hotspots in central downtown.

Map 1: Downtown-YSU-Mercy Health Shuttle Routes



Sources: Streetlight data, 2019

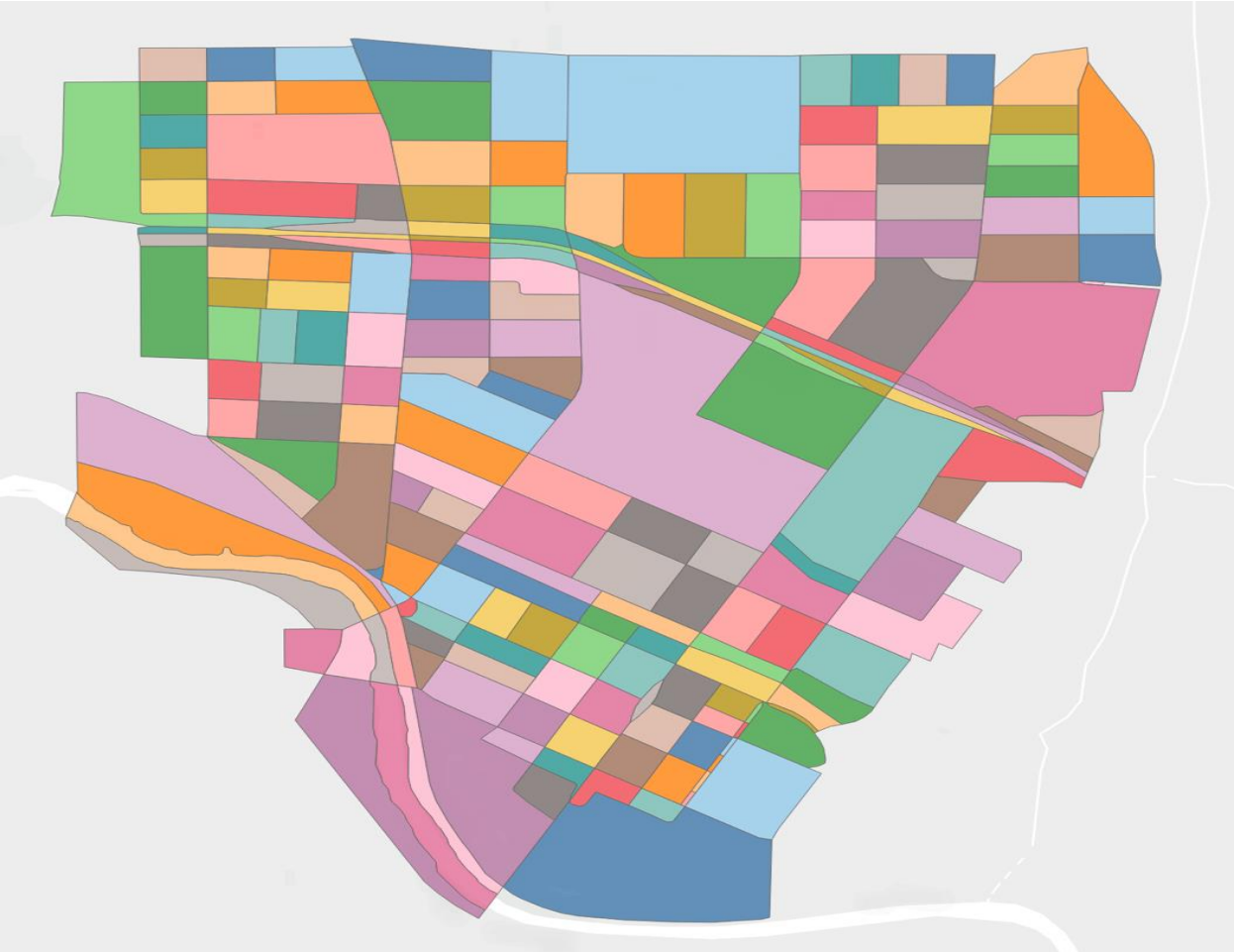
To perfect the proposed route and better understand its impact on the Youngstown population, Western Reserve Transit Authority and Eastgate partnered with Youngstown State

University to create a study of the proposed route as well as the general traffic patterns throughout Youngstown. The beginning of work began in September of 2021 when the Youngstown State Economics department created a survey to analyze the interest in the autonomous shuttle vehicle in Youngstown. The survey showed promising results, with over 85% of respondents being interested in trying a trip in an autonomous vehicle with a safety operator. Following the survey, YSU faculty from the departments of Mathematics and Statistics, Geography, Economics, and Computer Science joined to create a special project class where three students were invited to use Streetlight's October 2019 Youngstown Traffic GPS data to analyze the traffic patterns throughout Youngstown. The research will serve to better understand the region's traffic patterns throughout key geographical zones, inform city planners of demographic areas of interest within the region, evaluate the proposed route for the autonomous shuttle, and provide valuable insights to business owners about the daily habits of the population.

Data

The data that were used for this study was provided by Streetlight. Streetlight is a private company that has licensed its geospatial traffic data to the Ohio Department of Transportation to be used in city planning efforts. Eastgate Regional Council of Governments, in partnership with the Western Reserve Transit Authority, gained access to these data due to its status as a Metropolitan Transportation Organization. The Streetlight data set receives over forty billion anonymized location records from smartphones and navigation devices in connected cars and trucks per month. These data are then indexed and processed through Streetlight's machine learning algorithms to provide a digestible and completely anonymous dataset of traffic within specified regions. The data gathered for the Youngstown Streetlight project was for the month of October 2019. This was the chosen time span, as traffic after this point was unrepresentative of typical traffic patterns in Youngstown due to the COVID-19 pandemic. Additionally, upon the advice of our industry experts and Youngstown State University faculty, the month of October is the best representative of typical University traffic, as classes are in session, the weather is fair, and there are no major holidays to disrupt typical traffic patterns. The geospatial data consists of an index of 194 distinct origins and destinations corresponding to 194 census blocks throughout the Youngstown area. In addition to geospatial data, the Streetlight dataset also contains three tables with over one hundred unique fields and approximately seven million total records corresponding to a unique origin-destination path for every time and day during the week. The time periods are broken down into seven individual days, as well as one additional classification for the entire week. Additionally, the data is broken down further into twenty-four individual time spans, with each category containing one hour of the day.

Map 2: 194 Youngstown Census Blocks



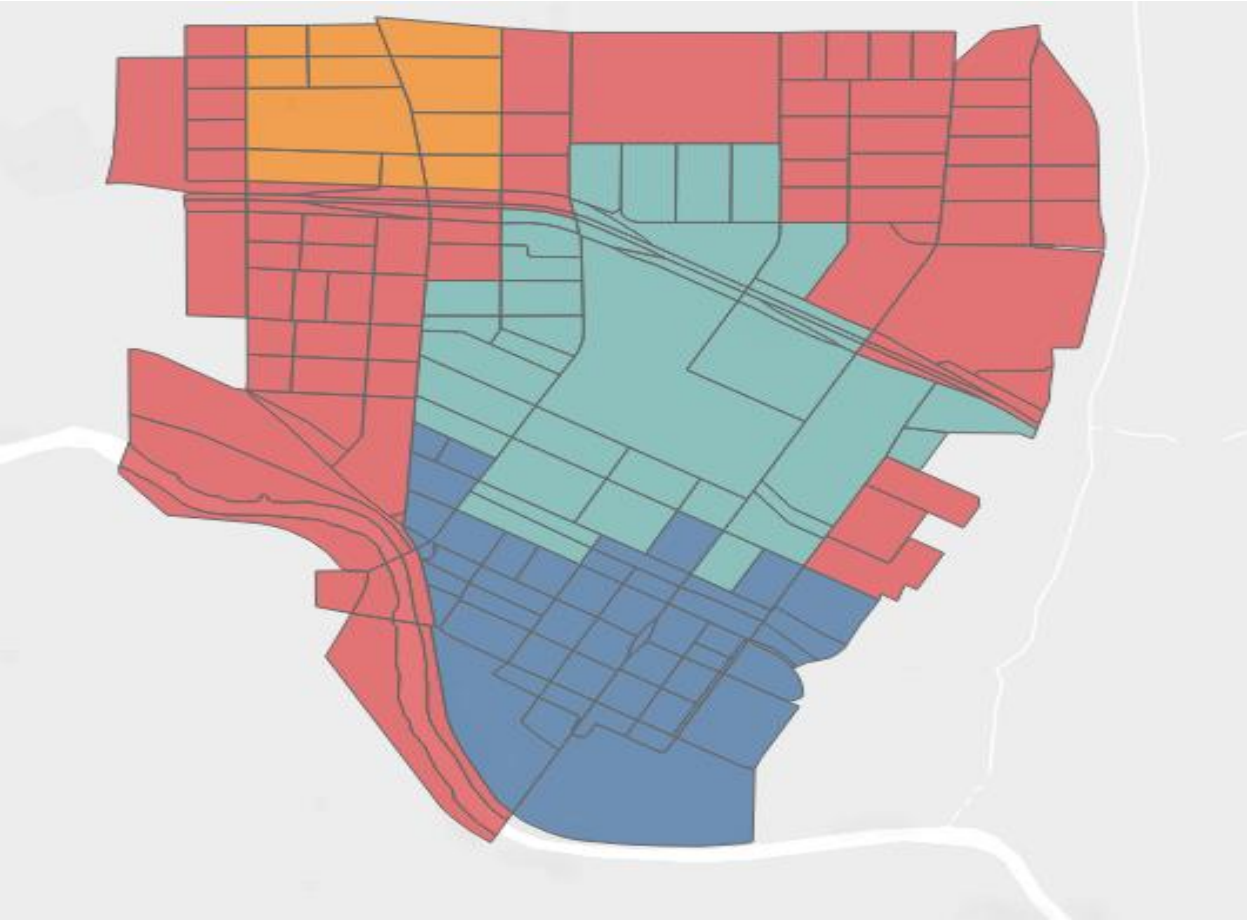
Sources: Streetlight data, 2019

Map 2 shows all the 194 zones originally contained in the dataset. To begin our analysis of the data, much work was done to mold these tables into one workable database. The primary task was to convert the individual origin and destination tags, consisting of census block IDs, into sensible groupings of relevant geographical destinations across Youngstown.

Our first grouping aggregated census blocks together into four key zones: the orange zone which defines Mercy Health, the number one employer and the largest hospital in the area, the teal zone which defines the entirety of Youngstown State University Campus, the blue zone which

defines the main part of Youngstown's Downtown district, and the red zone which defines the areas outside of Youngstown's three key zones. A visual depiction of these groupings can be seen in Map 3.

Map 3: 4 Youngstown Regions

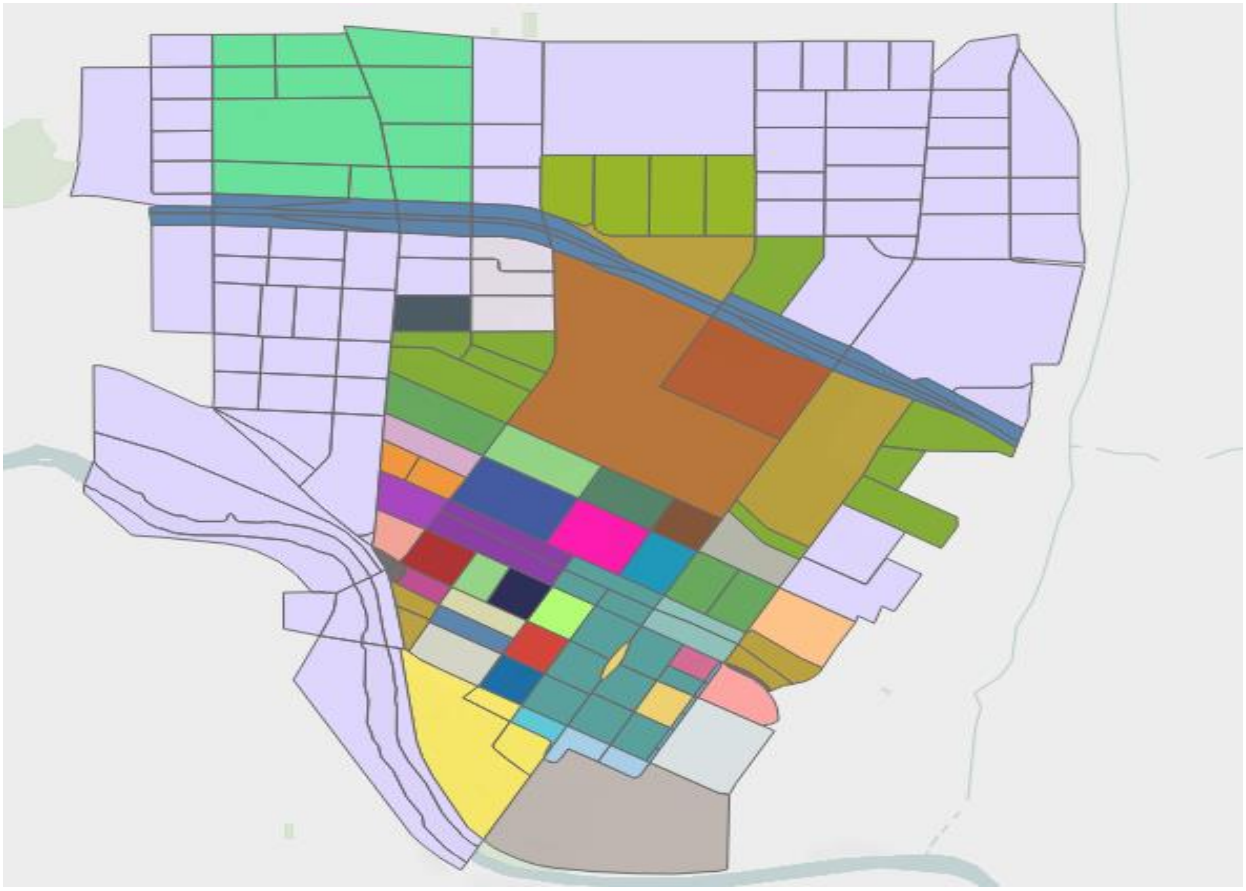


Sources: Streetlight data, 2019

After this initial grouping, we then began further dividing the regions into distinct location zones, representing specific geographical areas of interest to the general Youngstown population. In addition to these location zones, location metadata was assigned to each zone signifying the features of the location: such as if it contains a restaurant or other relevant buildings. To assist this

grouping, we consulted with a local industry expert, Charlie Nelson, and geography faculty from Youngstown State University. Together, we were able to accurately group and identify these relevant locations, to capture a representative sample of traffic to and from relevant areas across Youngstown. During the consultation, the fourth zone that defines the area outside of the three main zones was removed from the study as time prohibited us from studying this area – this large area is colored in light purple in Map 4. Given more time, this area could easily be further explored. Some key features of the excluded zone include large areas of severe poverty, greater than all other locations within the study area. Additionally, the excluded zone contains an area far from the proposed shuttle route, and analysis can be done to anticipate the impact of those travelers farther from the autonomous shuttle.

Map 4: Initial Location Groupings



Sources: Streetlight data, 2019

With this nearly completed geography, we began to run into recurring problems regarding the block size. The large block represented by a light brown color in Map 4 is the center of Youngstown State University and contained too many important college buildings necessary to study. As a result, when attempting to analyze traffic from a specific location located within this block, we could only capture the average traffic going to all buildings within this large block. A similar issue occurred on the dark yellow block located adjacent to the large YSU block. To solve this problem, a new geographical set of locations was created, altering our grouping by census blocks in just these two key areas. The geography was divided up instead by building location, and

with this final grouping shown in Map 5, we were then able to analyze traffic to and from all college buildings and all housing on campus individually.

Map 5: Finalized Location Groupings – Broken Up Blocks

Sources: Streetlight data, 2019

Following our final location grouping, we then began creating a uniform dataset that combines all relevant variables across the three tables provided in the Streetlight dataset. Upon completion of our data cleaning, we were left with fifty-four distinct fields, with 6,794,250 rows.

Some of the key variables in our final dataset included the *Average Daily Traffic from an Origin to a Destination, Average Travel Time from an Origin to a Destination, Average Daily Total Origin Traffic from an Origin, Average Daily Total Destination Traffic to a Destination, Trip Purpose (Home-Based Other, Home-Based Work, Non-Home Based),* and a variety of socio-economic variables.

Summary Statistics

Table 1. Descriptive Statistics for Individual Travelers (By Educational Level)

Variables	***Average In Percent***
Educational Level	
Graduate Degree	6.69%
Bachelor's Degree	11.87%
No High School Diploma	14.83%
Some College	25.20%
High School Diploma	41.42%
Total	***100.00%***

Sources: Streetlight data, 2019

Table 1 shows the summary statistics for individual travelers categorized by their education level in the Youngstown area. The summary shows that the highest traveler in this category is the High School Diploma holders of about 42% whilst the lowest traveler is the Graduate degree holders of about 7%.

Table 2. Descriptive Statistics for Individual Travelers (By Race)

Variable	***Average In Percent***
Race	
Islander	0.03%
Indian	0.25%
Asian	0.84%
Other Race	1.59%
Multiple Races	2.63%
Hispanic	4.73%
Black	20.68%
White	69.25%
Total	***100.00%***

Sources: Streetlight data, 2019

Table 2 shows the summary statistics for individual travelers categorized by their Race in the Youngstown area. The summary shows that the highest traveler in this category is the white race of about 70% whilst the lowest traveler is an Islander of about 0.03%.

Table 3. Descriptive Statistics for Individual Travelers (By Income Level)

Variables	***Average in Percent***
Income Level	
Income More than 200K	1.30%
Income 150K to 200K	1.72%
Income 125K to 150K	2.37%
Income 100K to 125K	4.62%
Income 75K to 100K	8.84%
Income 35K to 50K	13.88%
Income 50K to 75K	16.51%
Income 20K to 35K	20.52%
Income Less than 20K	30.24%
Total	***100.00%***

Sources: Streetlight data, 2019

Table 3 shows the summary statistics for individual travelers categorized by their level of income in the Youngstown area. The summary shows that the highest traveler in this category is the individuals with an income of less than $20,000 about 31% whilst the lowest traveler is the individuals with an income of more than $200,000 about 2%.

Table 4. Descriptive Statistics for Individual Travelers (With or Without Kids)

Variables	***Average in Percent***
Individuals with or without kids	
With Kids under 6 years	12.67%
With Kids between 6-17 years	13.53%
With Kids	24.00%
With No Kids	49.99%
Total	**100.19%**

Sources: Streetlight data, 2019

Table 4 shows the summary statistics for individual travelers categorized by whether the individual is with or without kids in the Youngstown area. The summary shows that the highest traveler in this category is the individual with no kids at about 50% whilst the lowest traveler is the individual with kids under 6 years of about 13%.

Analysis

With the final dataset created, elementary analysis began to identify the changes in traffic patterns and demographics between the four large regional areas. To begin, the focus of our analysis was strictly on the traffic volume between the distinct origin-destination pairs, designated by a unique time, day, origin, and destination location.

Fig 1. Downtown-YSU-Mercy Health Matrix

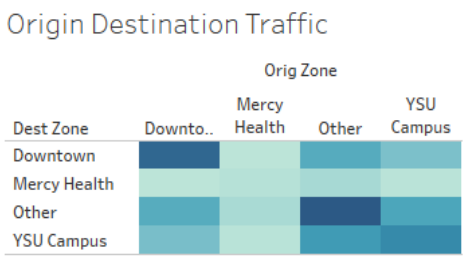
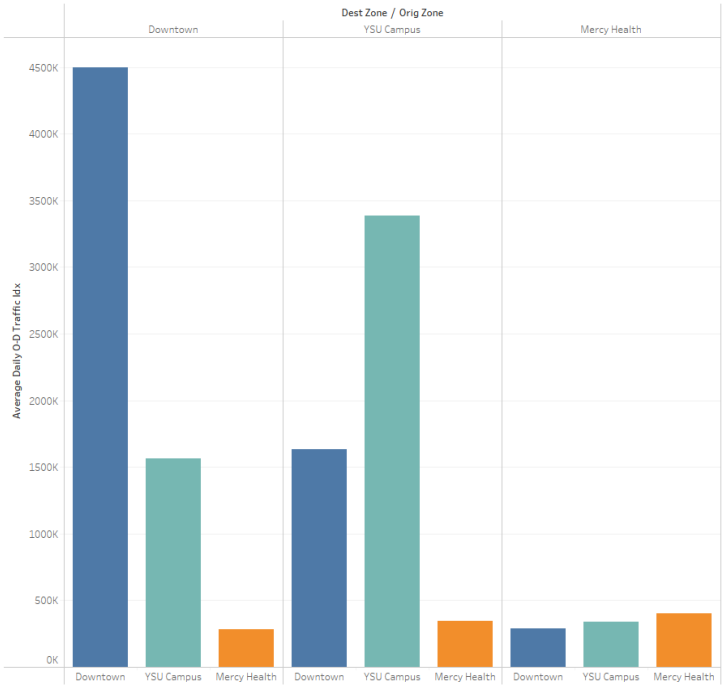


Fig 2. Downtown-YSU-Mercy Health Bar Graph



Sources: Streetlight data, 2019

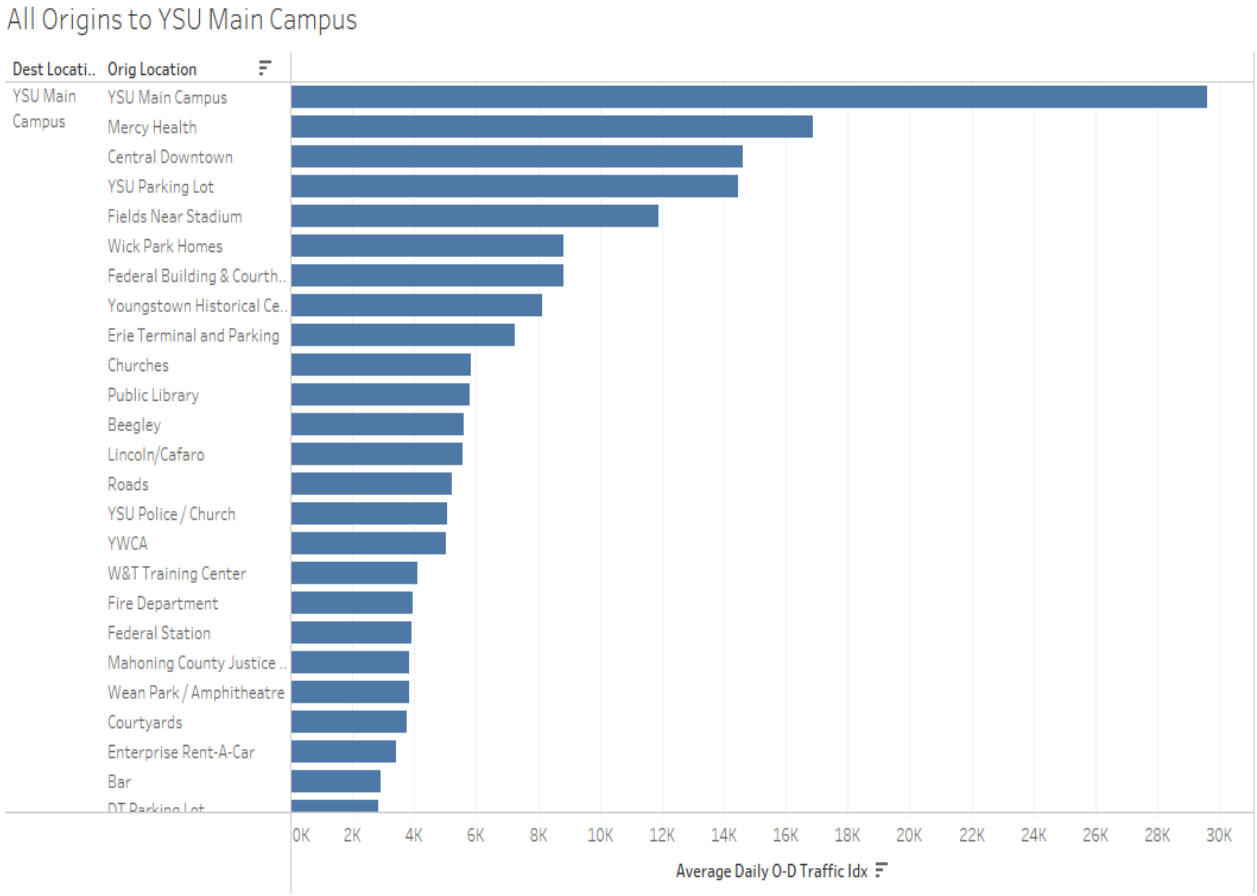
Upon initial analysis, we noted that intrazonal traffic – traffic beginning and ending in the same origin zone – dominated the traffic for every origin-destination zone pair. This trend is captured clearly in **Figures 1 & 2**, analyzing the traffic between origin and destination regions within Youngstown.

Traffic originating in the downtown zone and ending in the downtown zone makes up most of the traffic within our three key regions. This is followed by traffic originating at the Youngstown State University campus and ending within the YSU campus. When looking at traffic from all three zones to Mercy Health, we see that the difference in traffic between the origin locations is much less severe than in the other two regions. We believe that this is because Mercy Health hospital attracts all types of people, and unlike downtown Youngstown and YSU, it does not attract a young population interested in recreational activities.

Upon further investigation of the traffic patterns between the three key regions, we realized that more in-depth analysis was required to answer more specific questions. For example, we were unable to track traffic patterns between intra-regional blocks, we were only able to analyze traffic from one major region to another.

As a result, a further breakdown of the regions began, and analysis began on the individual locations within the three key regions of Youngstown. Again, traffic volume was the initial focus of understanding the individual locations comprising the three main regions. One approach to understanding the traffic to and from locations in Youngstown was to look at the most popular origin locations and destinations relating to one specific location.

Fig 3. All Origins of YSU Main Campus



Sources: Streetlight data, 2019

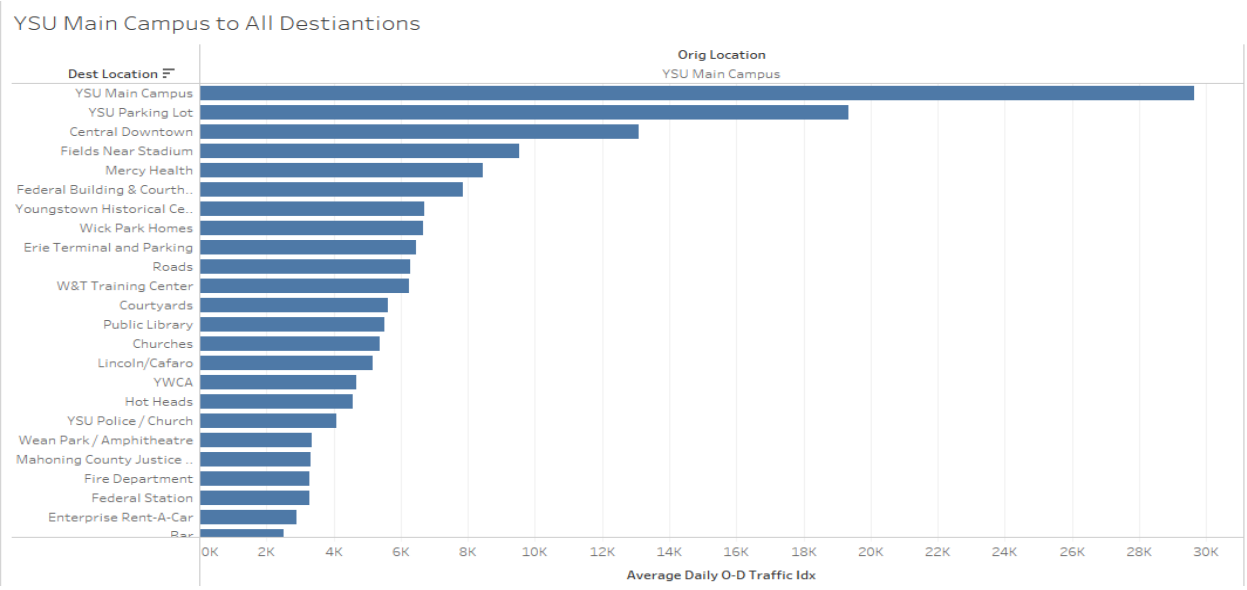
By utilizing our location groupings, we were now able to quickly identify origin and destination hotspots across Youngstown. In **Figures 3 & 4**, we examine the traffic from all locations across Youngstown to the center of campus at YSU, and all traffic originating at YSU to all locations across Youngstown (only the first few locations are shown, to increase clarity). Additionally, both "other" and "422" were removed from the possible location, as due to the size and traffic counts of these areas, the results for the relevant locations were obscured. When analyzing **Figures 3 & 4**, we can see that traffic from Youngstown State University's main campus is to other locations

near campus: intrazonal traffic back to the main campus, parking lots around the university, and the fields near Stambaugh stadium. Alongside these main destinations are Central Downtown and Mercy Health, the only two non-YSU locations in the top five destinations.

When looking at the graph reporting traffic from all locations to YSU's main campus, we see a similar story with most of the traffic being intrazonal, and the only two non-YSU locations making up the origin traffic being Central Downtown and Mercy Health.

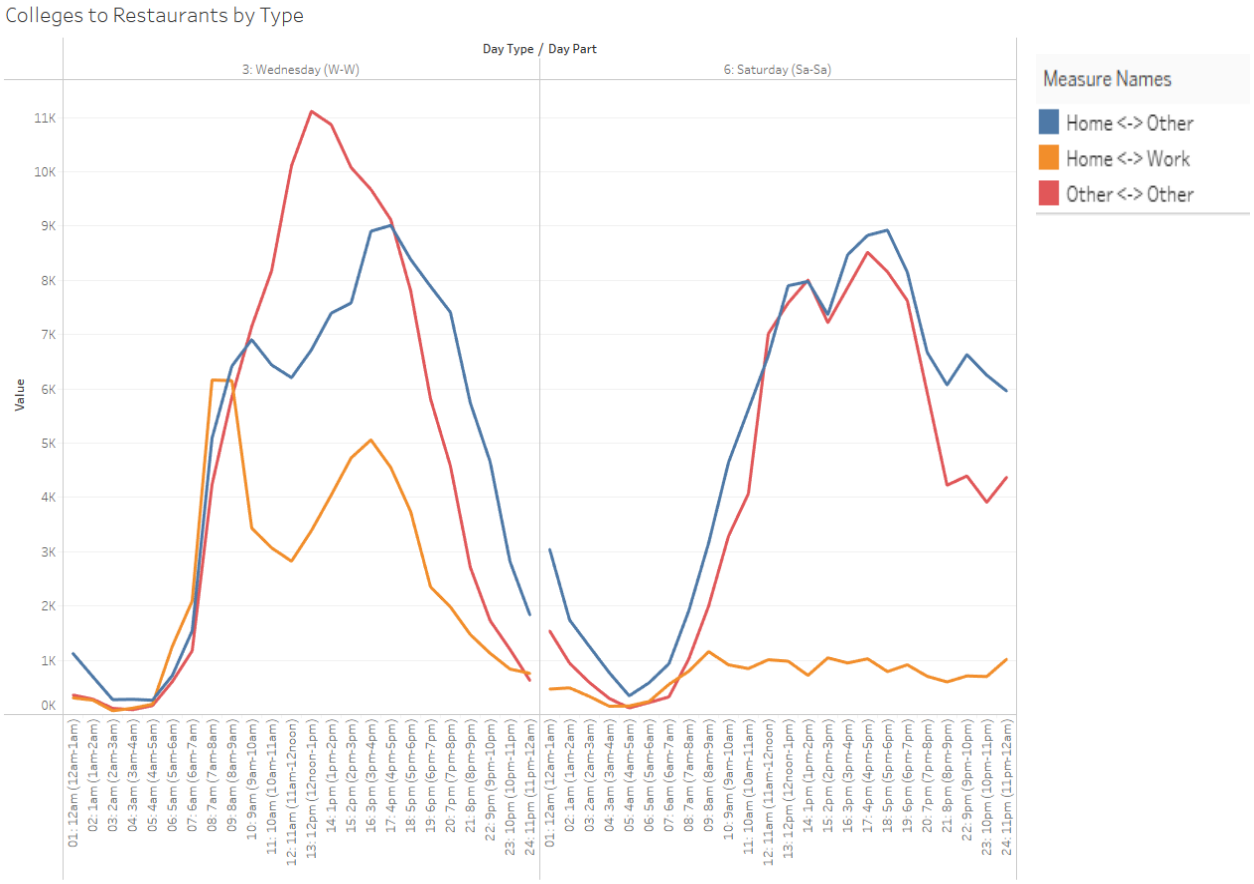
Although the analysis of traffic to and from YSU is the only traffic pattern shown here, we can analyze the traffic for all origin and destination location combinations, and filter by time and day of the trip. Furthermore, by utilizing the metadata associated with the individual locations, one can quickly group destination or origin traffic by relevant properties like containing a restaurant, containing a parking lot, or containing housing.

Fig 4. YSU Main Campus to All Destinations



Sources: Streetlight data, 2019

Fig 5. All YSU Colleges to Restaurant Traffic



Sources: Streetlight data, 2019

An example of this higher-level analysis is shown in **Figure 5**, where the focus of the visualization is to explain the traffic of YSU students to Youngstown-based restaurants. To capture the student population, a subset of the total locations containing colleges on campus was identified and used as the origin. To map to all restaurants, all blocks containing metadata associated with restaurants were grouped and designated as the destination. Further dimensions were added to the visualization, including daypart, daytime, and types of traffic. In **Figure 5**, we can see the blue line representing traffic to/from a home location to/from a non-work location, the orange line representing traffic to/from a home location to/from a non-work location, and the red line representing traffic to/from a non-home/work location to/from a non-home/work location.

In **Figure 5,** we can see distinct differences between the type of traffic occurring throughout the day on Wednesday and Saturday. One key difference is that Wednesday traffic sees an early morning peak occurring between 6 and 8 am, not seen on Saturday traffic. This is accounted for by the students not working on weekends, as the primary source of traffic at this time on Wednesday is home-based work traffic, which is not at all relevant on Saturday. Another difference is the drop-off of late-night restaurant traffic after 6 pm. On Wednesday, we see a sharp drop-off, due to classes and work that typically occurs on a Thursday. This trend does not appear to be as significant on Saturday, as traffic after 6 pm and up until 2 pm, is much more popular compared to Wednesday. This is again likely due to the lack of classes and works on a Sunday, allowing students to stay out late at bars and restaurants until early Sunday morning.

Traffic analysis between destination origin location pairs is not the only type of analysis that can be done with the Streetlight data. We can also analyze the *type of people* making these trips between locations, however, due to time constraints not much work has been done yet.

Further Analysis using the dashboard

With the combination of variables, the three datasets have, there are a lot of analyses/ visualizations that can be done. We have only included a few of the key analyses in this report.

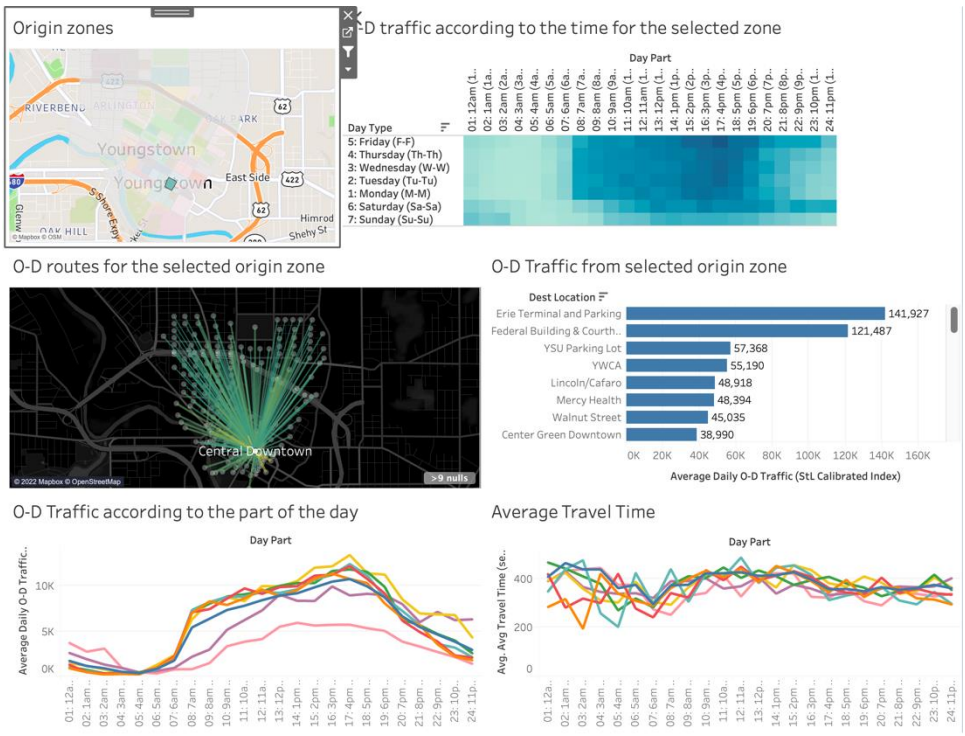


Figure 6: An example of a Tableau Dashboard

However, for readers who would like to do a specific analysis, we have created a Tableau dashboard that allows users to query between the different variables we have.

The figure above shows a Tableau Dashboard. On the top left part of the figure, we can select the origin zone (Central Downtown for example). Then rest of the visualizations in the dashboard will change incorporating Central Downtown as the origin zone. This allows us to see the traffic from Central Downtown to the rest of the destination zones. The time heat map (top right) shows the traffic for various parts of the day through the week for Central Downtown as the origin. The O-D routes show the traffic going from Central Downtown to the rest of the zones. We can also see

the traffic according to different days of the week, the average travel time, and the top destinations for trips originating from Central Downtown. For visualizations for a specific day, we can select the desired day on the top right part, and the rest of the visualizations will be filtered for that day. This is one example of a dashboard we have created. For more information about the tools that we have used and some data visualization techniques, refer to the Appendix section.

Conclusion

In summary, the highest individual travelers, categorized by their level of education are the individuals with High School Diploma holders of about 42%, by their race, the White race emerged as the highest travelers of about 70%, by their level of income, the individuals with an income less than $20,000 emerged as the highest travelers of about 31% and by having a kid or not, the individuals with no kids emerged as the highest travelers of about 50%.

Travel patterns are primarily analyzed using traffic count/index and travel time. In this project, the average traffic index was used for most of the analysis. We started by looking at the origin and destination locations that had the highest traffic, which allowed us to study the key travel spots. After that, analysis was done on different days of the week. As discussed in the dashboard section, there are different combinations of analysis that can be done. We spent more than half of the semester understanding the data and experimenting with different data methods, however, we did have some key takeaways. One notable feature of all traffic in our dataset was that intrazonal traffic dominated all the possible origin-destination pairs. Additionally, traffic frequently peaked between 6-8 am, 11-1 pm, and 4-6 pm. However, when diving up the data by day, we see that these peaks did not occur at 6-8 am and 11-1 pm. One cause of this discrepancy was due to the reduced work-

based travel that takes place on weekends. Additionally, weekend traffic featured a significantly higher 9 pm-3 am travel activity compared to weekdays. This heightened late night and early morning activity may also be an influence on the lack of 6-8 am and 11-1 pm peaks.

The dataset we were provided with has a lot of information and potential for more advanced analysis. We did not analyze all the variables at our disposal, but future students who take on this or similar projects can utilize the data for advanced analysis. From here on, specific advanced analysis can be performed. For example, businesses interested in customer traffic can focus on the traffic to their location of interest and conduct the analysis. The foundations that we got from the DATAX courses (Predictive Modeling, Data Visualization, and Data Management) immensely helped us through the project, and this course, DATAX 5986 is an effective installment for the students who took these three courses. We were able to utilize the skills we had and work on a real work project which had real-world use. We have added an appendix section that explains our data methods discussing some of the visualizations we have used with the goal of providing resources to future students who will take this class or similar project. For interested readers, all our works including the dashboards can be accessed online through proper authorization from Dr. Kerns.

Appendix

The goal of the appendix section is to explain some of the data analysis methodologies that we have used in this project. We will discuss how and why we did some of the visualizations in Tableau. The main sections of this report will be more analysis and result-focused, but we also wanted to add this appendix section for the readers who might want to reproduce our work or do further analysis.

A. Software used

While most of the work was done using Tableau, we used various other software thorough the project. Before diving into Tableau, we started the initial data cleaning in R. We used R to get information about different variables in the three datasets we had. Some of the initial joins were also done using R but later we migrated to Tableau as it provided a smoother and streamlined process for analysis. Within Tableau, we used Tableau Prep

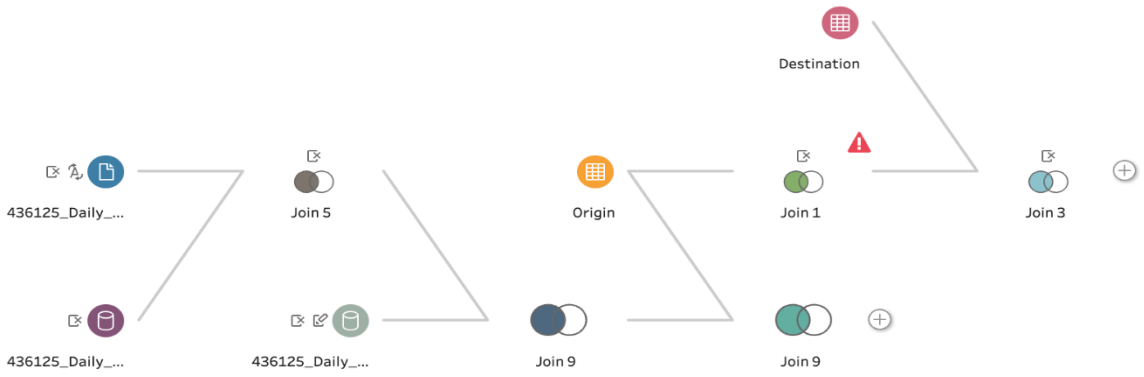


Figure 7: Example of a Tableau Prep Builder Project

Builder to clean and join our datasets. Regardless of whether you want to use Tableau for your data analysis, Tableau Prep Builder is an excellent tool to prepare the data for analysis. It lets you see every step of data cleaning/joining so that other team members can easily contribute without confusion and repetition as shown in figure 7.

The datasets we worked with contained a geospatial aspect. This means we had to analyze the data with respect to the locations. We were also provided shapefiles that store the spatial data like Excel worksheets store tabular data. Tableau lets you import shapefiles, but the functions are limited. For some advanced functions regarding shapefiles, we used ArcGIS. In the shapefile, the geometry for zones was polygon, but to create links between the origin and destination points, we needed the longitude and latitude of the zones. Therefore, the conversion from polygon to points (as longitude and latitude of the centroid of the polygon) was done using ArcGIS. We were then able to use the points to create links in Tableau which will be discussed below. Using ArcGIS, we can do advanced analyses like finding the shortest route between two zones. Although our goal for the project did not include finding routes between the zones, future students who have taken on such projects can incorporate it. Lastly, ArcGIS Insight was also used to visualize the route links which are shown below.

B. Creating route links between origin and destination points.

The best way to visualize the traffic from an origin zone is using link maps. This allows us to see the distribution of traffic from one origin zone to various destinations.

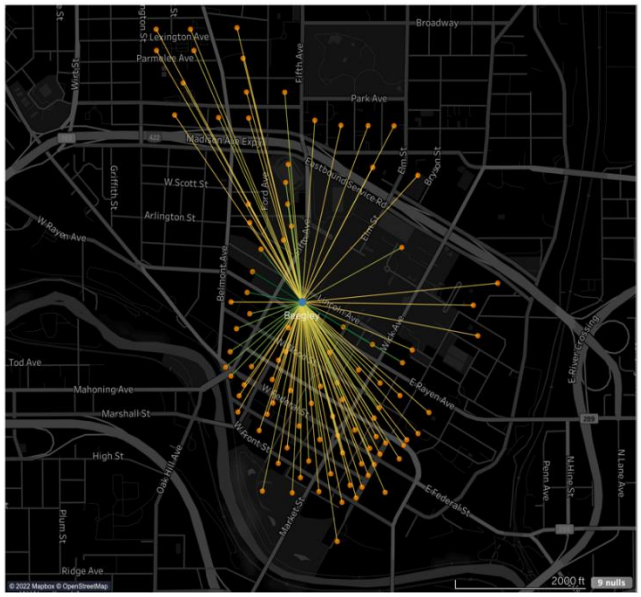


Figure 8: Route links created using Tableau

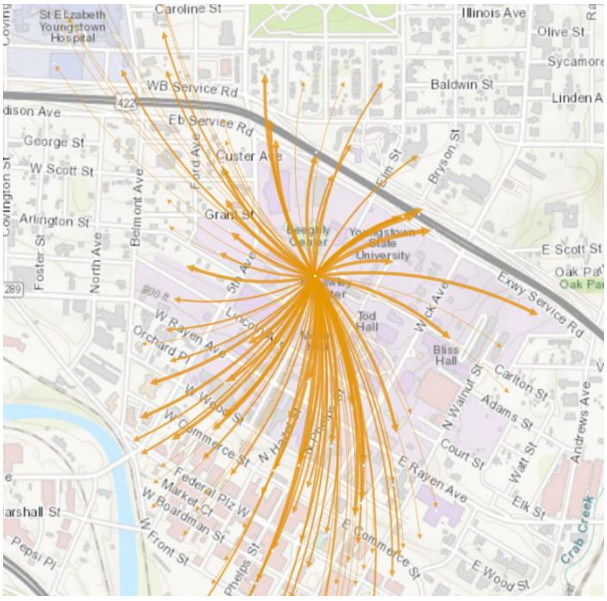


Figure 9: Route links created using ArcGIS Insight

We can also visualize traffic from multiple origin zone to multiple destination zones. We can see the visualization of the route links in figures 8 and 9. While the initial visualizations were done using ArcGIS insight, we also used Tableau to create dashboards containing the link maps. To create these maps, we need to use the coordinates of the origin and destination point in form of longitude and latitude. As mentioned above the longitude and latitude for the zones were extracted from ArcGIS. We join the coordinate data with the traffic dataset in Tableau. Now we use the 'Calculated Field' option in Tableau to create the sets of origin and destination points. We create a calculated field using 'Make Point' using the longitude and latitude (or x and y coordinates sometimess) for both origin and destinations naming them 'Origin Point' and 'Destination Point.' Once we have these two fields, we need to create a route between the points. For this, create a calculated field 'Routes' using the 'Make Line' option where the starting point is 'Origin Point,'

and the endpoint is ‘Destination Point.’ Then drag ‘Routes’ to the worksheet to visualize the routes. We can change the color of the routes according to different variables. For example, in the picture shown above, the color of the routes depends on the origin-destination traffic, green meaning higher traffic.

C. Origin-Destination Matrix

Origin-Destination(O-D) matrix represents the movement of people from one location to another. Different traffic patterns can be extracted using the O-D matrix. We can use several measurements to build an O-D matrix. For example, traffic count, traffic index, travel time, etc. For this project,

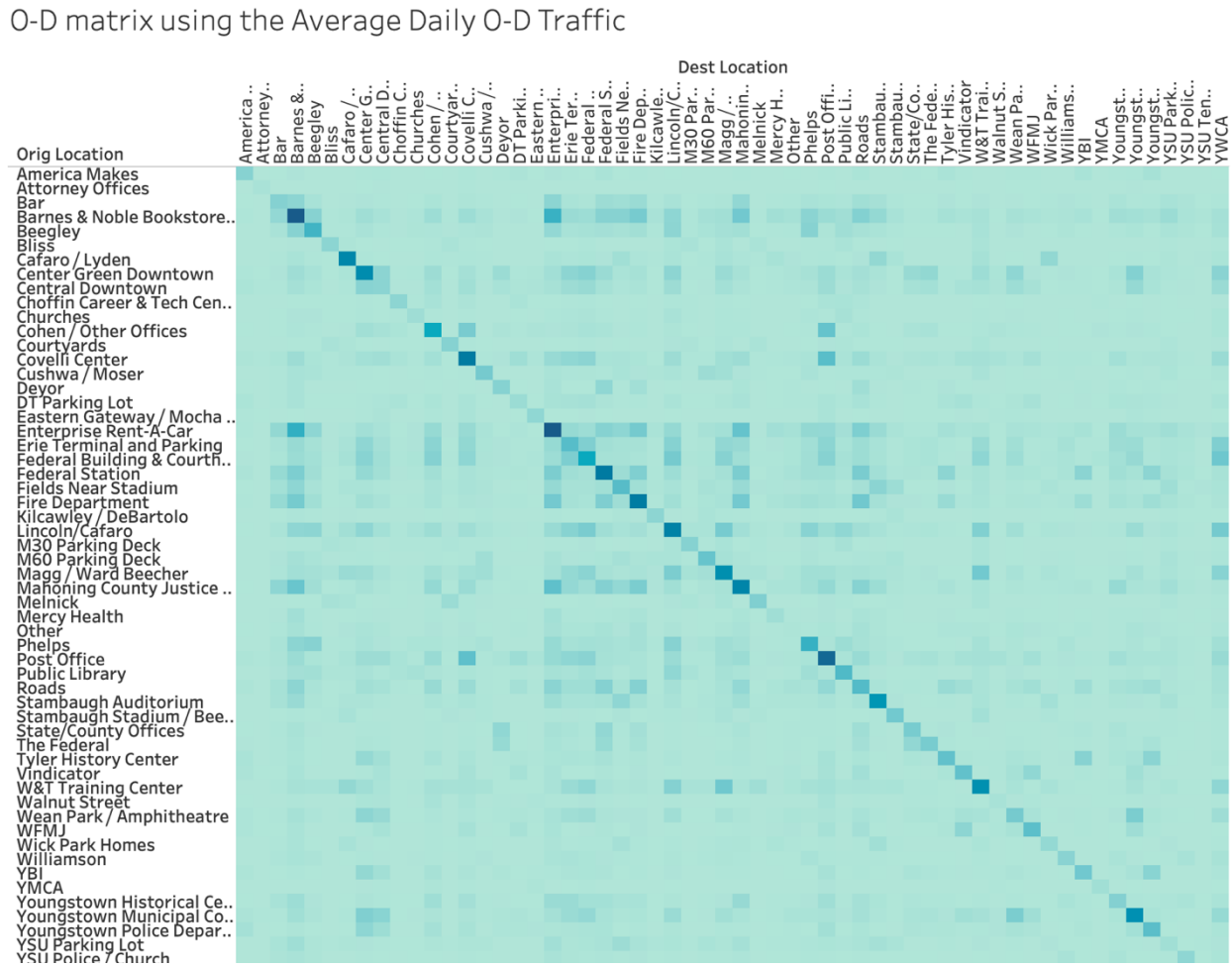
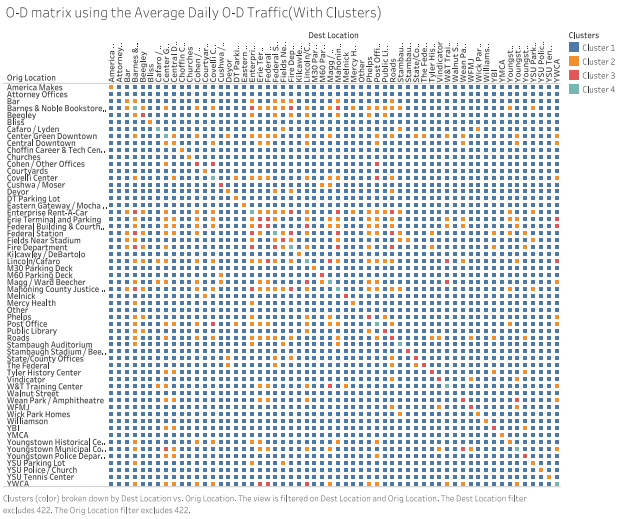


Figure 10: O-D matrix

we primarily focused on the daily average origin-destination traffic index for creating the O-D matrix as shown in Figure 10.

O-D matrix is an effective way to visualize the overall traffic. From the O-D matrix shown above, from the diagonal entries, we can see that the intrazonal traffic is much higher than the interzonal traffic. O-D matrix is a base for more advanced analysis to extract the traffic patterns. We can filter the matrix for various parts of the days and see how the traffic changes. To group the locations with similar traffic, we can use the clustering feature Tableau has. Tableau uses K-means clustering where we can choose the variables and numbers of clusters. The figure below shows the Cluster model applied to the O-D matrix where the variable used was the O-D traffic. There are four clusters, and the figure below shows the description of the clusters.



Inputs for Clustering

Variables:	Avg. Average Daily O-D Traffic (StL Calibrated Index)
Level of Detail:	Dest Location, Orig Location
Scaling:	Normalized

Summary Diagnostics

Number of Clusters:	4
Number of Points:	3364
Between-group Sum of Squares:	12.625
Within-group Sum of Squares:	1.3531
Total Sum of Squares:	13.978

		Centers
Clusters	**Number of Items**	**Avg. Average Daily O-D Traffic (StL Calibrated Index)**
Cluster 1	2945	25.255
Cluster 2	342	174.64
Cluster 3	62	477.54
Cluster 4	15	1430.0
Not Clustered	0	

Figure 11: Clustering using the O-D matrix

We can use more than one variable to model our clusters. For example, if we want to cluster the trips with similar traffic and travel time together, we can do this by using the traffic and travel time variable to cluster the O-D matrix.